INCH-POUND

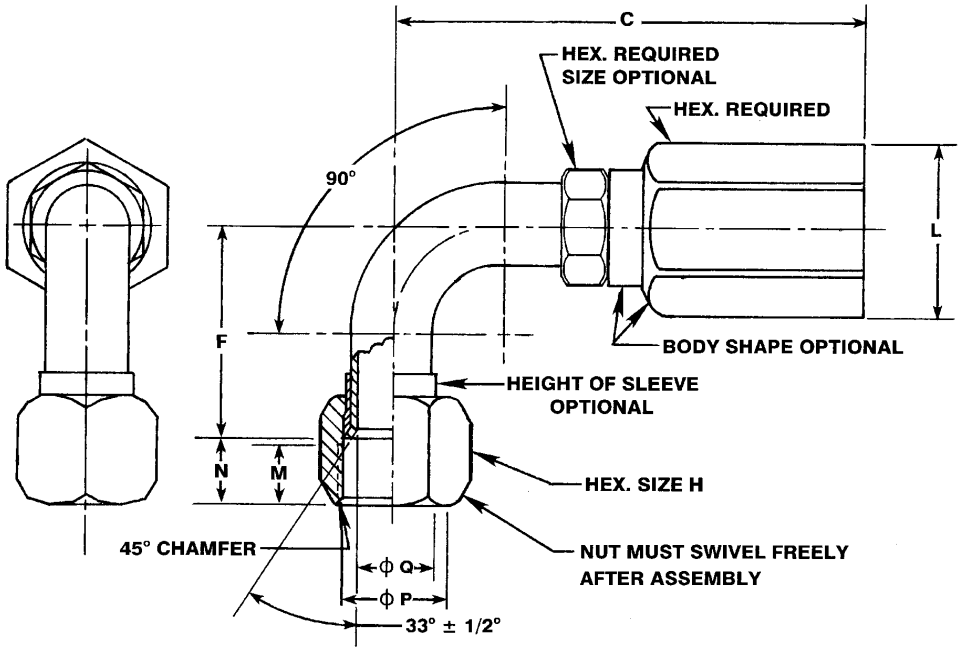
MIL-DTL-52525/7B
30 June 1998
SUPERSEDING
MIL-F-52525/7A
30 June 1987

DETAIL SPECIFICATION SHEET

# FITTINGS, HOSE, SCREW-ON TO 37° FLARE FEMALE SWIVEL, 90° BENT TUBE, SHORT DROP, TYPE 100R1

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-52525. The fittings described herein are for use with MIL-DTL-52471, type 100R1, single wire braid hose.




NOTES:
1. DIMENSION P TO BE CONCENTRIC WITH DIMENSION Q WITHIN 0.015 FIM
2. Q=BORE DIAMETER

FIGURE 1. Fitting, screw-on to 37° flare, 90° bent tube, short drop.



AMSC N/A FSC 4730

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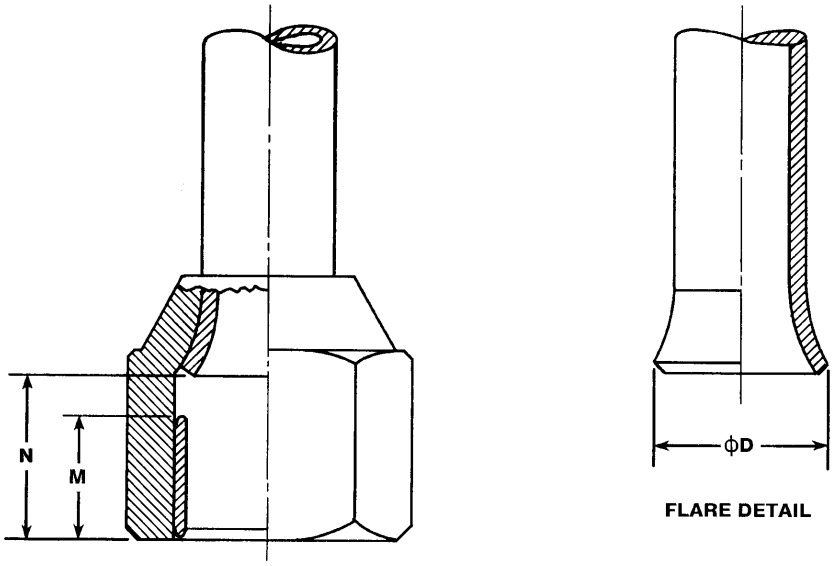


FIGURE 2. Alternate termination radial flare.

TABLE I. Dimensions for type 100R1 fittings. 1/

| Part number | Hose dash no. | Hose I.D. | Tube dash no. | Tube O.D. | Thread size | C Maximum | F Maximum | H Hex size |
|---|---|---|---|---|---|---|---|---|
| M52525/7-4-4 | 4 | 1/4 | 4 | 1/4 | 7/16-20 | 2.63 | 0.74 | 9/16 |
| M52525/7-4-5 | 4 | 1/4 | 5 | 5/16 | 1/2-20 | 2.67 | 0.83 | 5/8 |
| M52525/7-6-6 | 6 | 3/8 | 6 | 3/8 | 9/16-18 | 3.03 | 0.91 | 11/16 |
| M52525/7-8-8 | 8 | 1/2 | 8 | 1/2 | 3/4-16 | 3.56 | 1.15 | 7/8 |
| M52525/7-8-10 | 8 | 1/2 | 10 | 5/8 | 7/8-14 | 3.78 | 1.29 | 1 |
| M52525/7-12-12 | 12 | 3/4 | 12 | 3/4 | 1-1/16-12 | 4.28 | 1.88 | 1-1/4 |
| M52525/7-16-16 | 16 | 1 | 16 | 1 | 1-5/16-12 | 5.20 | 2.20 | 1-1/2 |

1/ All dimensions not shown shall conform to SAE J516.



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TABLE I. Dimensions for type 100R1 fittings - Continued. 1/

| Part number | L Maximum | M Minimum | N +0.030 −0.000 | P ±0.005 | Q Minimum |
|---|---|---|---|---|---|
| M52525/7-4-4 | 0.87 | 0.300 | 0.344 | 0.295 | 0.11 |
| M52525/7-4-5 | 0.87 | 0.331 | 0.375 | 0.355 | 0.11 |
| M52525/7-6-6 | 1.09 | 0.333 | 0.375 | 0.435 | 0.26 |
| M52525/7-8-8 | 1.20 | 0.375 | 0.422 | 0.570 | 0.38 |
| M52525/7-8-10 | 1.20 | 0.461 | 0.500 | 0.680 | 0.38 |
| M52525/7-12-12 | 1.59 | 0.469 | 0.562 | 0.850 | 0.57 |
| M52525/7-16-16 | 2.02 | 0.563 | 0.594 | 1.000 | 0.78 |

1/ All dimensions not shown shall conform to SAE J516.

TABLE II. Dimensions for alternate termination. 1/

| Tube size | D | | M Minimum | N Minimum |
|---|---|---|---|---|
| | max | min | | |
| 1/4 | 0.360 | 0.340 | 0.295 | 0.375 |
| 5/16 | 0.430 | 0.400 | 0.295 | 0.375 |
| 3/8 | 0.490 | 0.460 | 0.297 | 0.385 |
| 1/2 | 0.660 | 0.630 | 0.339 | 0.438 |
| 5/8 | 0.790 | 0.760 | 0.409 | 0.521 |
| 3/4 | 0.950 | 0.920 | 0.464 | 0.594 |
| 1 | 1.200 | 1.170 | 0.495 | 0.625 |

1/ Dimensions C, F, H, and Q shall be in accordance with table I.



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TABLE III. Maximum operating pressures of fittings.

| Hose I.D. | PSI |
|---|---|
| 1/4 | 2750 |
| 5/16 | 2500 |
| 3/8 | 2250 |
| 1/2 | 2000 |
| 5/8 | 1500 |
| 3/4 | 1250 |
| 1 | 1000 |

TABLE IV. Over-tightening torque for swivel nut.

| Tube O.D. | Torque |
|---|---|
| | inch pounds (in-lb) |
| 1/4 | 135 |
| 5/16 | 180 |
| 3/8 | 270 |
| 1/2 | 450 |
| 5/8 | 650 |
| 3/4 | 900 |
| 1 | 1200 |

REQUIREMENTS:

1. Fittings shall be accordance with figure 1 and table I, III, and IV. The use of alternate termination flare as shown in figure 2 and corresponding dimensions of table II is at the option of the contractor.

2. Materials and finishes shall be as specified in MIL-DTL-52525.

3. Marking shall include the manufacturer’s name or trademark, size, and hose identifier R1. Location of marking is optional.



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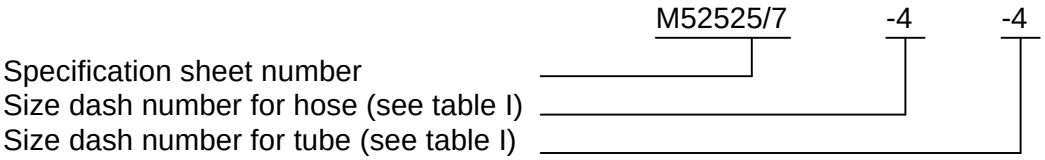
NOTES:

1. Dimensions are in inches.

2. Part number.

Example of a Military part number: M52525/7-4-4




## CONCLUDING MATERIAL

Custodian:
Army - AT
Air Force - 99

Preparing activity:
DLA - CC

(Project 4730-0859-07)

Review activities:
Air Force - 82
Army - AR



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